DSP · MOCK

DSP mock A

Answers and worked solutions · Day 22 · Sat 14 Nov · pack DX1

Marking guide

Mark yourself honestly. Give method marks where the method is right and only the arithmetic slips. Give no marks for a correct number with no working on a part worth 2 or more marks.

Question 1 [8 marks]

(a) The highest frequency is $F_{\max} = 1100\pi/(2\pi) = 550$ Hz. Nyquist rate $= 2F_{\max} = \mathbf{1100}$ **Hz**. $F_s = 800$ Hz is below this, so the 550 Hz component will alias. The 150 Hz component is fine.

(b) $T = 1/800$ s. Each component has $f = F/F_s$:

- 150 Hz: $f = 150/800 = 0.1875$, so $\omega = 2\pi(0.1875) = 0.375\pi$. Already in range.
- 550 Hz: $f = 550/800 = 0.6875$, so $\omega = 1.375\pi$. Subtract 2π: $\omega = -0.625\pi$.

So $2\sin(1.375\pi n) = 2\sin(-0.625\pi n) = -2\sin(0.625\pi n)$, and

$$\boxed{x(n) = 3\cos(0.375\pi n) - 2\sin(0.625\pi n)}$$

(c) The ideal D/A puts each principal frequency back at $F = fF_s$: $0.1875 \times 800 = 150$ Hz and $0.3125 \times 800 = 250$ Hz.

$$\boxed{y_a(t) = 3\cos(300\pi t) - 2\sin(500\pi t)}$$

The 550 Hz tone comes back as a 250 Hz tone with its sign flipped.

(d) $y(n) = x(2n) = x_a(2nT)$. This is the same as sampling at 400 Hz.

$$y(n) = 3\cos(0.75\pi n) - 2\sin(1.25\pi n).$$

$1.25\pi - 2\pi = -0.75\pi$, so $-2\sin(1.25\pi n) = -2\sin(-0.75\pi n) = +2\sin(0.75\pi n)$. Both components now sit at $\omega_0 = 0.75\pi$:

$$y(n) = 3\cos(0.75\pi n) + 2\sin(0.75\pi n).$$

Use $a\cos\theta + b\sin\theta = \sqrt{a^2+b^2}\cos(\theta - \tan^{-1}(b/a))$:

$$\boxed{y(n) = 3.606\cos(0.75\pi n - 0.588)}, \quad A = \sqrt{13} = 3.606,\ \omega_0 = 0.75\pi,\ \phi = -0.588 \text{ rad } (-33.7^\circ)$$

(e)

System	Linear	Time invariant	Causal	Static	BIBO stable
(i) $x(n) - 0.5n\,x(n-1)$	Yes	**No**: coefficient n changes with time	Yes	**No**: uses $x(n-1)$	**No**: $\lvert x\rvert \le 1$ gives $\lvert y\rvert$ growing like $0.5n$
(ii) $\lvert x(n+1)\rvert$	**No**: $\lvert -x\rvert \neq -\lvert x\rvert$	Yes	**No**: uses future input $x(n+1)$	**No**: depends on a sample other than $x(n)$	Yes: $\lvert y\rvert = \lvert x\rvert \le B$

- a. 1 mark: 1100 Hz and “not adequate, 550 Hz aliases”.
- b. 2 marks: 1 for 0.375π, 1 for $-2\sin(0.625\pi n)$ with the sign right. Half credit for $+2\sin(0.625\pi n)$.
- c. 1 mark: 250 Hz with the minus sign.
- d. 2 marks: 1 for showing both terms at 0.75π, 1 for A and ϕ.
- e. 2 marks: 1 per system, all five properties right.

COMMON MISTAKES

- Folding a sine without flipping its sign. Cosine is even, sine is odd: $\sin(-\theta) = -\sin\theta$.
- Writing $\omega = 2\pi F$ instead of $\omega = 2\pi F/F_s$.
- In (d), treating $x(2n)$ as halving the frequency. Discarding samples *doubles* every digital frequency, which is why aliasing gets worse.
- Calling (ii) static. Memoryless means $y(n)$ depends on $x(n)$ only. $x(n+1)$ is not $x(n)$.
- Calling (i) stable because it has no feedback. The growing coefficient n makes it unstable.

REREAD: D1 (sampling, aliasing, system properties).

Question 2 [7 marks]

(a) Characteristic equation: $\lambda^2 - \frac{1}{4}\lambda - \frac{1}{8} = 0$, so $(\lambda - \frac{1}{2})(\lambda + \frac{1}{4}) = 0$.

$$\lambda_1 = \tfrac{1}{2},\ \lambda_2 = -\tfrac{1}{4}; \qquad \text{modes } \left(\tfrac{1}{2}\right)^n,\ \left(-\tfrac{1}{4}\right)^n$$

Both $|\lambda| < 1$ and the system is causal, so it is **BIBO stable**.

(b) $y(n) = \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) + x(n) + x(n-1)$, with $x(0..3) = 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}$.

$$\begin{aligned}
y(0) &= 1 \\
y(1) &= \tfrac{1}{4}(1) + \tfrac{1}{3} + 1 = \tfrac{19}{12} = 1.5833 \\
y(2) &= \tfrac{1}{4}\cdot\tfrac{19}{12} + \tfrac{1}{8}(1) + \tfrac{1}{9} + \tfrac{1}{3} = \tfrac{139}{144} = 0.9653 \\
y(3) &= \tfrac{1}{4}\cdot\tfrac{139}{144} + \tfrac{1}{8}\cdot\tfrac{19}{12} + \tfrac{1}{27} + \tfrac{1}{9} = \tfrac{1015}{1728} = 0.5874
\end{aligned}$$

(c) For $n \geq 1$ the input $\delta(n) + \delta(n-1)$ is zero, so $h(n) = c_1(\frac{1}{2})^n + c_2(-\frac{1}{4})^n$ for $n \geq 1$ (and also at $n = 0$ here). Get the initial values by recursion: $h(0) = 1$, $h(1) = \frac{1}{4}(1) + 1 = 1.25$.

$$\begin{aligned} c_1 + c_2 &= 1 \\ \tfrac{1}{2}c_1 - \tfrac{1}{4}c_2 &= 1.25 \end{aligned} \quad \Rightarrow \quad c_1 = 2,\ c_2 = -1$$

Check by partial fractions: $H(z) = \dfrac{1 + z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})} = \dfrac{2}{1 - \frac{1}{2}z^{-1}} - \dfrac{1}{1 + \frac{1}{4}z^{-1}}$. Then

$$\boxed{h(n) = \left[2\left(\tfrac{1}{2}\right)^n - \left(-\tfrac{1}{4}\right)^n\right]u(n)}$$

Check: $h(2) = 0.5 - 0.0625 = 0.4375$; recursion gives $\frac{1}{4}(1.25) + \frac{1}{8}(1) = 0.4375$.

(d) Use $a^n u(n) * b^n u(n) = \dfrac{a^{n+1} - b^{n+1}}{a - b}u(n)$.

$$2\left(\tfrac{1}{2}\right)^n * \left(\tfrac{1}{3}\right)^n = 2\,\frac{(\frac{1}{2})^{n+1} - (\frac{1}{3})^{n+1}}{1/6} = 6\left(\tfrac{1}{2}\right)^n - 4\left(\tfrac{1}{3}\right)^n$$

$$\left(-\tfrac{1}{4}\right)^n * \left(\tfrac{1}{3}\right)^n = \frac{(-\frac{1}{4})^{n+1} - (\frac{1}{3})^{n+1}}{-7/12} = \tfrac{3}{7}\left(-\tfrac{1}{4}\right)^n + \tfrac{4}{7}\left(\tfrac{1}{3}\right)^n$$

Subtract the second from the first:

$$\boxed{y(n) = \left[6\left(\tfrac{1}{2}\right)^n - \tfrac{3}{7}\left(-\tfrac{1}{4}\right)^n - \tfrac{32}{7}\left(\tfrac{1}{3}\right)^n\right]u(n)}$$

Check: $y(0) = 6 - \frac{3}{7} - \frac{32}{7} = 1$. $y(1) = 3 + \frac{3}{28} - \frac{32}{21} = 1.5833$. Both match part (b).

- a. 1 mark: both modes and a stability statement.
- b. 2 marks: 1 for the recursion set up correctly, 1 for all four values.
- c. 2 marks: 1 for the form and initial values (or the partial fractions), 1 for c_1, c_2.
- d. 2 marks: 1 for the geometric-convolution formula used twice, 1 for the final coefficients.

COMMON MISTAKES

- Using $h(0) = 1$ and $h(1) = \frac{1}{4}$, i.e. forgetting the $x(n-1)$ term. The $+\delta(n-1)$ adds 1 to $h(1)$.
- Getting the characteristic equation signs wrong. Move the y terms to the form $\lambda^2 - \frac{1}{4}\lambda - \frac{1}{8}$ exactly as they appear on the left-hand side.
- In (d), dropping the $u(n)$, or forgetting that $(-\frac{1}{4}) - \frac{1}{3} = -\frac{7}{12}$ is negative.
- Not checking (d) against (b). Two lines of checking catch most errors in this question.

REREAD: D2 (LCCDE, modes, $h(n)$, convolution).

Question 3 [7 marks]

(a) $h(n) = \{\underset{\uparrow}{0.5}, 1, 0.5\}$.

$$H(\omega) = 0.5 + e^{-j\omega} + 0.5e^{-j2\omega} = e^{-j\omega}\left(0.5e^{j\omega} + 1 + 0.5e^{-j\omega}\right) = \boxed{(1+\cos\omega)\,e^{-j\omega}}$$

Since $1 + \cos\omega \geq 0$ for all ω:

$$|H(\omega)| = 1 + \cos\omega, \qquad \theta(\omega) = -\omega, \quad -\pi < \omega < \pi$$

There are no $\pm\pi$ jumps (the amplitude never goes negative). $|H(0)| = 2$ and $|H(\pi)| = 0$, so this is a lowpass filter.

(b) Yes. $\theta(\omega) = -\omega n_0$ with $n_0 = 1$, so the group delay is $\tau(\omega) = -\frac{d\theta}{d\omega} = \mathbf{1\ sample}$. This agrees with $h(n)$ being symmetric about $n = 1$.

(c) Parseval: $\displaystyle\int_{-\pi}^{\pi} |H(\omega)|^2 d\omega = 2\pi \sum_n |h(n)|^2 = 2\pi(0.25 + 1 + 0.25) = \boxed{3\pi}$.

(d) Use $A\cos(\omega_0 n + \phi) \to A|H(\omega_0)|\cos(\omega_0 n + \phi + \theta(\omega_0))$ for each term.

Term	ω_0	$\lvert H\rvert$	θ	Output
4	0	2	0	8
$2\cos(\frac{\pi}{2}n + \frac{\pi}{6})$	$\pi/2$	1	$-\pi/2$	$2\cos(\frac{\pi}{2}n - \frac{\pi}{3})$
$3\cos(\frac{2\pi}{3}n)$	$2\pi/3$	0.5	$-2\pi/3$	$1.5\cos(\frac{2\pi}{3}n - \frac{2\pi}{3})$
$5(-1)^n = 5\cos(\pi n)$	π	0	(not needed)	0

$$\boxed{y(n) = 8 + 2\cos\left(\tfrac{\pi}{2}n - \tfrac{\pi}{3}\right) + 1.5\cos\left(\tfrac{2\pi}{3}n - \tfrac{2\pi}{3}\right)}$$

Because the phase is linear, every surviving term is scaled by its gain and delayed by exactly one sample. A simulation of the difference equation matches this exactly for $n \geq 2$.

- a. 2 marks: 1 for $H(\omega)$ factored, 1 for magnitude and phase.
- b. 1 mark: "yes, 1 sample".
- c. 1 mark: 3π with Parseval named.
- d. 3 marks: 1 for the DC term, 1 for the two cosines, 1 for recognising the $(-1)^n$ term is removed.

COMMON MISTAKES

- Leaving $(-1)^n$ out of the frequency analysis. It is a sinusoid at $\omega = \pi$.
- Forgetting to multiply the DC term by $H(0) = 2$.
- Writing Parseval without the 2π: $\int_{-\pi}^{\pi} |H|^2 d\omega = 2\pi \sum |h|^2$, not $\sum |h|^2$.
- Adding the phase to the amplitude, or using degrees and radians in the same expression.

REREAD: D4 (frequency response, sinusoidal steady state). For part (c), D3 (DTFT properties, Parseval).

Question 4 [8 marks]

(a) Lengths 4 and 3, so the result has length $4 + 3 - 1 = 6$.

$$x_1 * x_2 = \{\underset{\uparrow}{2}, 5, 3, 0, -1, -1\}$$

(b) 4-point circular convolution = linear convolution wrapped modulo 4. Add samples 4 and 5 onto samples 0 and 1:

$$x_1 \circledast_4 x_2 = \{2 + (-1),\ 5 + (-1),\ 3,\ 0\} = \boxed{\{\underset{\uparrow}{1}, 4, 3, 0\}}$$

Circular equals linear when $N \geq L_1 + L_2 - 1 = \mathbf{6}$.

(c) For $N = 4$, $W_4 = e^{-j\pi/2} = -j$, so $X_1(k) = \sum_{n=0}^{3} x_1(n)(-j)^{kn}$.

$$\begin{aligned} X_1(0) &= 1 + 2 + 0 - 1 = 2 \\ X_1(1) &= 1 + 2(-j) + 0 - 1(j) = 1 - 3j \\ X_1(2) &= 1 - 2 + 0 - 1(-1) = 0 \\ X_1(3) &= 1 + 2(j) + 0 - 1(-j) = 1 + 3j \end{aligned}$$

$$\boxed{X_1(k) = \{2,\ 1 - 3j,\ 0,\ 1 + 3j\}}$$

Check: $X_1(3) = X_1^*(1)$, as it must be for a real sequence.

(d) Resolution $= F_s/N \leq 30$ Hz, so $N \geq 8000/30 = 266.7$. The next power of 2 is $\boxed{N = 512}$ ($m = 9$).

$$T_0 = N/F_s = 512/8000 = \boxed{0.064\ \text{s} = 64\ \text{ms}}$$

(e) No. The true resolution is set by the record length: $1/T_0 = F_s/L = 8000/200 = 40$ Hz, which is wider than the 30 Hz tone spacing. Zero-padding only samples the same DTFT more finely. It does not add information, so the two peaks stay merged.

- a. 1 mark: correct six samples.
- b. 2 marks: 1 for the wrapped result, 1 for $N \geq 6$.
- c. 2 marks: 1 for $X_1(0), X_1(2)$; 1 for $X_1(1), X_1(3)$.
- d. 2 marks: 1 for $N = 512$, 1 for T_0.
- e. 1 mark: "no" with the 40 Hz reasoning.

COMMON MISTAKES

- Using $e^{+j2\pi kn/N}$ in the forward DFT. The forward DFT has the minus sign.
- Rounding N to 267 or 300. The question asks for $N = 2^m$.
- Believing that zero-padding improves the ability to separate tones. It gives a smoother picture of the same, smeared spectrum.
- In (b), wrapping the wrong end. Samples with index $n \geq N$ are added to index $n - N$.

REREAD: D5 (DFT, circular convolution) for (a)–(c). D6 (spectral analysis, resolution, zero-padding) for (d)–(e).

Question 5 [8 marks]

(a) Denominator: $1 + 1.5z^{-1} - z^{-2} = (1 - 0.5z^{-1})(1 + 2z^{-1})$.

$$H(z) = \frac{3 + z^{-1}}{(1 - 0.5z^{-1})(1 + 2z^{-1})} = \frac{3z\left(z + \frac{1}{3}\right)}{(z - 0.5)(z + 2)}$$

Zero: $z = -\frac{1}{3}$ (and one at $z = 0$ when written in positive powers). **Poles:** $z = 0.5$ and $z = -2$.

(b) Partial fractions in z^{-1}:

$$A = \left.\frac{3 + z^{-1}}{1 + 2z^{-1}}\right|_{z^{-1}=2} = \frac{5}{5} = 1, \qquad B = \left.\frac{3 + z^{-1}}{1 - 0.5z^{-1}}\right|_{z^{-1}=-1/2} = \frac{2.5}{1.25} = 2$$

$$H(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{2}{1 + 2z^{-1}}$$

ROC	$h(n)$	Type
$\lvert z\rvert > 2$	$[(0.5)^n + 2(-2)^n]u(n)$	causal, unstable
$0.5 < \lvert z\rvert < 2$	$(0.5)^n u(n) - 2(-2)^n u(-n-1)$	noncausal, stable
$\lvert z\rvert < 0.5$	$-[(0.5)^n + 2(-2)^n]u(-n-1)$	anticausal, unstable

(c) Stable requires the ROC to include the unit circle $|z| = 1$. Only $\mathbf{0.5 < |z| < 2}$ does. That ROC is an annulus, so $h(n)$ is two-sided: **noncausal**. A causal system would need the ROC outside the outermost pole ($|z| > 2$), which excludes the unit circle. So this system cannot be both causal and stable.

(d) Cross-multiply $Y(z)(1 + 1.5z^{-1} - z^{-2}) = X(z)(3 + z^{-1})$:

$$\boxed{y(n) + 1.5y(n-1) - y(n-2) = 3x(n) + x(n-1)}$$

(e) The ROC includes $|z| = 1$, so $H(\omega) = H(z)|_{z=e^{j\omega}}$ exists. At $\omega = \pi$, $z = -1$:

$$H(\pi) = \frac{3 - 1}{1 - 1.5 - 1} = \frac{2}{-1.5} = -\frac{4}{3}$$

$$\boxed{y(n) = -\tfrac{4}{3}\cos(\pi n) = \tfrac{4}{3}\cos(\pi n + \pi)}$$

A numerical two-sided convolution with the stable $h(n)$ gives $y(0) = -1.3333$, which agrees.

- a. 1 mark: both poles and the zero.
- b. 3 marks: 1 for A and B, 2 for the three ROC/$h(n)$ pairs (lose 1 per wrong pair, minimum 0).
- c. 2 marks: 1 for the ROC with reason, 1 for "noncausal" with reason.
- d. 1 mark: correct LCCDE.
- e. 1 mark: $-4/3$ and the output.

COMMON MISTAKES

- Using $-(a)^n u(-n-1)$ for a pole *inside* the ROC's inner edge. Rule: for each pole, if the ROC is outside it the term is right-sided $u(n)$; if the ROC is inside it the term is left-sided $-u(-n-1)$.
- Missing the minus sign on the left-sided term.
- Saying "stable because both poles are inside the unit circle". The pole at -2 is outside. Stability depends on the ROC, not just the pole positions.
- Evaluating H on the unit circle for the causal system (ROC $|z| > 2$). Its frequency response does not exist.

REREAD: D7 (z-transform and ROC), D8 (inverse z-transform, stability and causality). D9 (mixed practice) for extra drill.

Question 6 [7 marks]

(a) $\omega_0 = 2\pi \times 125/1000 = \boxed{\pi/4}$ rad, so $2\cos\omega_0 = \sqrt{2}$.

$$H(z) = b_0\left(1 - \sqrt{2}z^{-1} + z^{-2}\right)$$

At $\omega = \pi$, $e^{-j\pi} = -1$ and $e^{-j2\pi} = 1$:

$$1 = b_0\left(1 + \sqrt{2} + 1\right) \Rightarrow b_0 = \frac{1}{2+\sqrt{2}} = \boxed{0.2929}$$

$$H(z) = 0.2929\left(1 - 1.4142z^{-1} + z^{-2}\right)$$

(b) $H(\omega) = b_0 e^{-j\omega}(2\cos\omega - \sqrt{2})$, so $|H(\omega)| = b_0|2\cos\omega - \sqrt{2}|$.

- 0 Hz ($\omega = 0$): $|H| = 0.2929(2 - 1.4142) = 3 - 2\sqrt{2} = \mathbf{0.172}$ (-15.3 dB).
- 250 Hz ($\omega = \pi/2$): $|H| = 0.2929 \times 1.4142 = \sqrt{2} - 1 = \mathbf{0.414}$ (-7.7 dB).

The notch is very wide. A second-order FIR notch has no poles to sharpen it, so it also strongly attenuates frequencies well away from 125 Hz.

(c) No. The zeros $e^{\pm j\pi/4}$ lie **on** the unit circle. The inverse $1/H(z)$ would have poles on the unit circle, which is marginally stable at best (it is a sinusoidal oscillator, not BIBO stable). A stable causal inverse needs all zeros strictly inside the unit circle (minimum phase).

(d) The comb has M zeros at $z_k = e^{j2\pi k/(M+1)}$, $k = 1, \ldots, M$, i.e. at $F = kF_s/(M+1)$. For nulls at multiples of 200 Hz:

$$\frac{F_s}{M+1} = 200 \Rightarrow M + 1 = \frac{1200}{200} = 6 \Rightarrow \boxed{M = 5}$$

Nulls in $0 < F \leq 600$ Hz: $\boxed{200,\ 400,\ 600\ \text{Hz}}$ ($k = 1, 2, 3$; 600 Hz is $\omega = \pi$, the zero at $z = -1$).

(e) $h(n) = \frac{1}{6}$ for $0 \leq n \leq 5$ is symmetric, so it has linear phase with delay $M/2 = \mathbf{2.5}$ **samples** $= 2.5/1200 = \mathbf{2.08}$ **ms**. DC gain: $H(0) = \sum h(n) = 6 \times \frac{1}{6} = \mathbf{1}$.

- a. 2 marks: 1 for ω_0, 1 for b_0.
- b. 1 mark: both gains and a sensible comment.
- c. 1 mark: “no”, zeros on the unit circle.
- d. 2 marks: 1 for $M = 5$, 1 for the three frequencies.
- e. 1 mark: delay and DC gain.

COMMON MISTAKES

- Normalising at $\omega = 0$ instead of $\omega = \pi$ as asked. That gives $b_0 = 1/(2 - \sqrt{2}) = 1.707$.
- Using $\omega_0 = 2\pi F$ (forgetting to divide by F_s).
- In (d), choosing $M = 6$ because “6 = 1200/200”. The zero spacing is $F_s/(M + 1)$, not F_s/M.
- Forgetting the null at $F_s/2 = 600$ Hz. It is there whenever $M + 1$ is even.

REREAD: D10 (digital filters: notch, comb, minimum phase). D4 for evaluating $|H(\omega)|$.

Question 7 [9 marks]

(a) $\omega = 2\pi F/F_s$:

$$\omega_p = \frac{2\pi(1000)}{8000} = \frac{\pi}{4}, \qquad \omega_s = \frac{2\pi(2000)}{8000} = \frac{\pi}{2}$$

(b) $T = 1$, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$\Omega_p = 2\tan\frac{\pi}{8} = 0.8284 \text{ rad/s}, \qquad \Omega_s = 2\tan\frac{\pi}{4} = 2.0000 \text{ rad/s}$$

(c) $\delta_p = 10^{-3/20} = 0.7079$, $\delta_s = 10^{-15/20} = 0.1778$.

$$N = \frac{\log\left(\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}\right)}{2\log\left(\dfrac{\Omega_s}{\Omega_p}\right)} = \frac{\log(30.77)}{2\log(2.4142)} = 1.944$$

Round up: $\boxed{N = 2}$.

(d) Match the stopband exactly (as in T9 Q5; with the bilinear transform there is no aliasing, so exceeding the passband spec is the sensible choice):

$$\Omega_c = \Omega_s\left(\frac{\delta_s^2}{1-\delta_s^2}\right)^{\frac{1}{2N}} = 2\left(\frac{0.03162}{0.96838}\right)^{1/4} = \boxed{0.8502 \text{ rad/s}}$$

For $N = 2$ the stable poles are $s_k = \Omega_c e^{j(\pi/2 + \pi(2k+1)/4)}$, $k = 0, 1$, i.e. $\Omega_c e^{\pm j3\pi/4}$. Their product is Ω_c^2 and their sum is $-\sqrt{2}\Omega_c$:

$$H(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\,\Omega_c s + \Omega_c^2} = \boxed{\frac{0.7228}{s^2 + 1.2024s + 0.7228}}$$

(Matching the passband instead gives $\Omega_c = 0.8294$. That also meets the spec and earns full marks if stated.)

(e) With $\Omega_c = 0.85$: $\Omega_c^2 = 0.7225$ and $\sqrt{2}\Omega_c = 1.2021$. Substitute $s = 2\dfrac{1-z^{-1}}{1+z^{-1}}$ and multiply top and bottom by $(1+z^{-1})^2$:

$$H(z) = \frac{0.7225(1+z^{-1})^2}{4(1-z^{-1})^2 + 2(1.2021)(1-z^{-1})(1+z^{-1}) + 0.7225(1+z^{-1})^2}$$

Collect powers of z^{-1} in the denominator:

$$\begin{aligned} z^0 &: 4 + 2.4042 + 0.7225 = 7.1267 \\ z^{-1} &: -8 + 1.4450 = -6.5550 \\ z^{-2} &: 4 - 2.4042 + 0.7225 = 2.3183 \end{aligned}$$

$$H(z) = \frac{0.7225 + 1.4450z^{-1} + 0.7225z^{-2}}{7.1267 - 6.5550z^{-1} + 2.3183z^{-2}} = \boxed{\frac{0.1014 + 0.2028z^{-1} + 0.1014z^{-2}}{1 - 0.9198z^{-1} + 0.3253z^{-2}}}$$

Difference equation:

$$\boxed{y(n) = 0.9198\,y(n-1) - 0.3253\,y(n-2) + 0.1014\,x(n) + 0.2028\,x(n-1) + 0.1014\,x(n-2)}$$

Checks:

- $\omega = 0$ ($z = 1$): $H = \dfrac{0.1014 + 0.2028 + 0.1014}{1 - 0.9198 + 0.3253} = \dfrac{0.4056}{0.4055} \approx 1$. Correct for a Butterworth lowpass.
- $\omega = \pi$ ($z = -1$): numerator $= 0.1014(1-1)^2 = 0$, so $H(\pi) = 0$. The bilinear transform maps $s = \infty$ to $z = -1$.

The design gives -2.79 dB at ω_p (spec: no worse than -3 dB) and -15.0 dB at ω_s. Its poles are at $0.570\,e^{\pm j0.633}$, inside the unit circle.

- a. 1 mark: both ω_p and ω_s.
- b. 1 mark: both prewarped values.
- c. 2 marks: 1 for δ_p, δ_s in linear scale, 1 for $N = 2$ from 1.944.
- d. 2 marks: 1 for Ω_c, 1 for $H(s)$.
- e. 3 marks: 1 for the substitution set up, 1 for normalised $H(z)$, 1 for the difference equation with correct signs (the check is part of this mark).

COMMON MISTAKES

- Skipping prewarping and using $\Omega = \omega/T$. That gives N and Ω_c for the wrong filter.
- Using $\Omega = \tan(\omega/2)$ without the factor $2/T$. Either is fine only if you are consistent and use the same factor in the bilinear substitution.
- Putting dB values straight into the N formula. Convert to linear: $\delta = 10^{-A/20}$.
- Rounding $N = 1.944$ down to 1.
- Sign errors moving from $H(z)$ to the difference equation. Denominator $1 - 0.9198z^{-1}$ becomes $+0.9198\,y(n-1)$ on the right-hand side.
- Forgetting to normalise so that $a_0 = 1$.

REREAD: D11 (IIR design: bilinear transform and Butterworth).

Question 8 [6 marks]

(a) Inverse DTFT of the ideal lowpass with $\omega_c = \pi/4$:

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} e^{j\omega n}d\omega = \frac{\sin(\pi n/4)}{\pi n} = \boxed{\tfrac{1}{4}\operatorname{sinc}\left(\tfrac{\pi n}{4}\right)}, \qquad h_d(0) = \tfrac{1}{4}$$

(course definition $\operatorname{sinc} x = \sin x/x$).

(b) Shift by $(M-1)/2 = 3$ and window: $h(n) = h_d(n-3)\,w(n)$, $0 \le n \le 6$, with Hamming $w(n) = 0.54 - 0.46\cos(2\pi n/6)$.

n	$n-3$	$h_d(n-3)$	$w(n)$	$h(n)$
0	-3	0.0750	0.08	0.0060
1	-2	0.1592	0.31	0.0493
2	-1	0.2251	0.77	0.1733
3	0	0.2500	1.00	0.2500
4	1	0.2251	0.77	0.1733
5	2	0.1592	0.31	0.0493
6	3	0.0750	0.08	0.0060

$$\boxed{h(n) = \{\underset{\uparrow}{0.0060}, 0.0493, 0.1733, 0.2500, 0.1733, 0.0493, 0.0060\}}$$

(These agree with `scipy.signal.firwin(7, 0.25, window='hamming', scale=False)`.)

(c) Symmetric: $h(n) = h(6-n)$. Linear phase with group delay $(M-1)/2 = \mathbf{3}$ **samples**.

(d) $H(0) = \sum h(n) = \mathbf{0.7073}$ (-3.0 dB). It is not 1 because the window truncates and tapers $h_d(n)$. The ideal filter's DC gain $\sum_{n=-\infty}^{\infty} h_d(n) = 1$ only holds for the full infinite sequence. With $M = 7$ and a Hamming taper, most of that sum is cut off. To fix it, divide every coefficient by $\sum h(n)$, i.e. $h(n)/0.7073$.

- a. 1 mark: $h_d(n)$ including the $n = 0$ value.
- b. 3 marks: 1 for the shift by 3, 1 for the window values, 1 for the products.
- c. 1 mark: symmetric, 3 samples.
- d. 1 mark: 0.7073 with a valid reason and fix.

COMMON MISTAKES

- Forgetting the shift: using $h_d(n)$ for $0 \le n \le 6$ gives a non-symmetric, non-linear-phase filter.
- Using MATLAB's sinc definition ($\sin(\pi x)/(\pi x)$) with the course's argument. Check: $h_d(0)$ must equal $\omega_c/\pi = 0.25$.
- Using the Hanning formula (endpoints 0) instead of Hamming (endpoints 0.08).
- Using $\cos(2\pi n/M)$ instead of $\cos(2\pi n/(M-1))$ in the window.

REREAD: D12 (Chebyshev and FIR design).

Score interpretation

Add up your marks out of 60. Then look at *where* you lost them, because that matters more than the total.

Score	What it means	What to do in the next 3 days
51–60 (85%+)	Exam-ready. Your methods are reliable under time pressure.	Do mock B if available. Spend the rest on your reference sheet and on speed.
42–50 (70–84%)	Solid. A few method gaps or slips.	Reread the packs for the two questions where you lost the most marks. Redo those questions from a blank page without the solution.
30–41 (50–69%)	Pass level, but fragile. Probably one or two whole topics are weak.	Spend one study block on each weak topic (pack sections 2 and 3, then the new practice questions). Put the method steps on your reference sheet.
Below 30	Big gaps. Most likely in L05–L10, which carry 45 of the 60 marks.	Go in this order: D8 (inverse z), D11 (Butterworth), D4 (frequency response), D5 (DFT). These are mechanical and give the most marks per hour.

Diagnose by question. Use this to choose what to reread.

Lost 3 or more marks on	Likely gap	Reread
Q1	Aliasing and folding sines, or property tests	D1
Q2	Initial values of $h(n)$, geometric convolution	D2
Q3	Factoring $H(\omega)$, phase, Parseval	D3, D4
Q4	DFT sums, wrap-around, resolution vs zero-padding	D5, D6
Q5	Choosing left- or right-sided terms from the ROC	D7, D8, D9
Q6	$\omega = 2\pi F/F_s$, normalisation, comb zero spacing	D10
Q7	Prewarping, order formula, bilinear algebra	D11
Q8	Shift by $(M-1)/2$, window formula, sinc definition	D12

Timing check. If you did not finish, note which question you were on at 120 minutes. Running out of time on Q7 usually means too long was spent on Q1–Q2. Those are worth only 15 marks, so aim to finish them in 30 minutes. If you finished early but scored below 42, slow down and use the checks built into the questions (Q2(d) against Q2(b), $H(1)$ and $H(-1)$ in Q7, conjugate symmetry in Q4(c)).